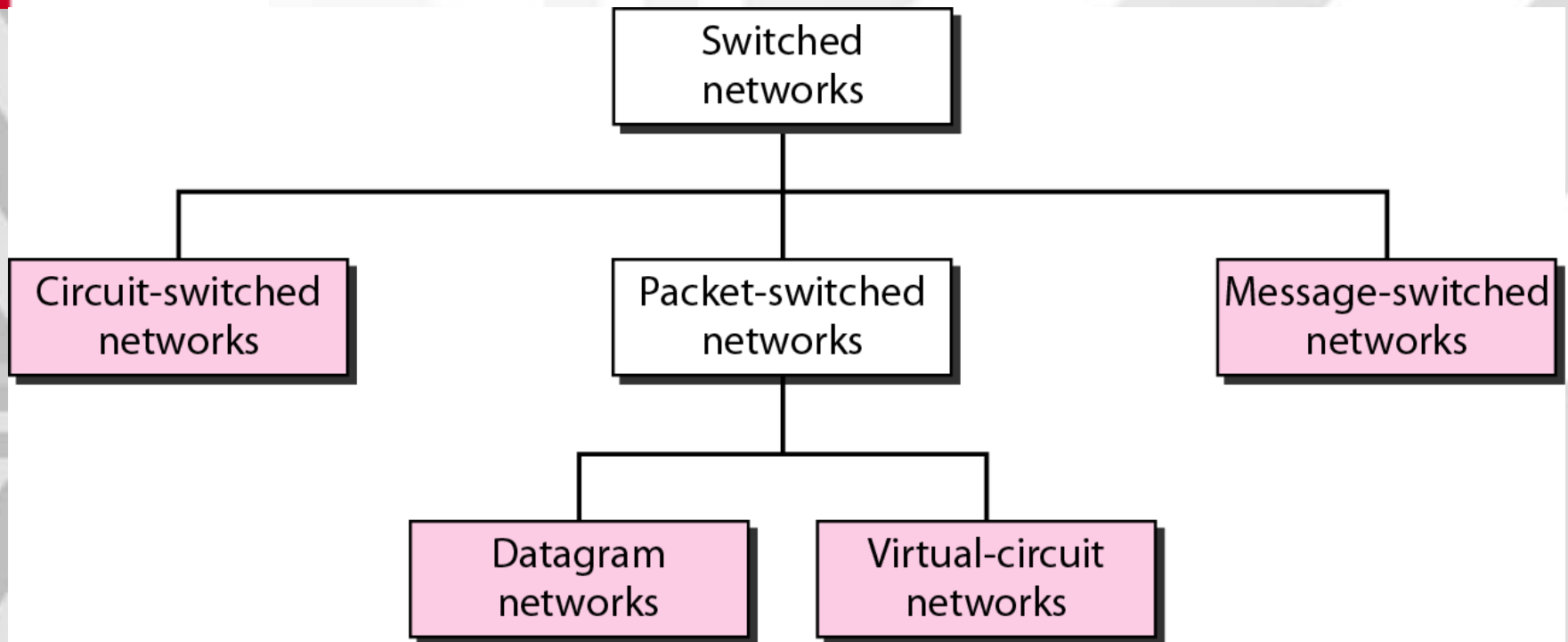


Computer Network and Communication CNS3200

Learning Outcome

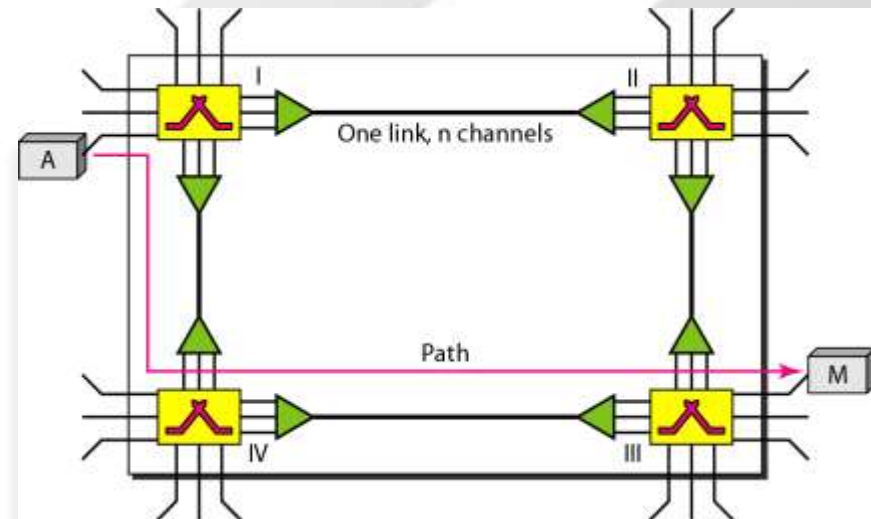
- **EXPLAIN AND CALCULATE MULTIPLEXING METHOD OF FREQUENCY DIVISION MULTIPLEXING (FDM) AND WAVELENGTH DIVISION MULTIPLEXING (WDM) (C4, NS)**
- **EXPLAIN THE METHODS OF GUIDED & UNGUIDED MEDIA TRANSMISSION (C4)**
- **DIFFERENTIATE PACKET AND CIRCUIT SWITCHING (C4)**

Taxonomy of switched networks



CIRCUIT-SWITCHED NETWORKS

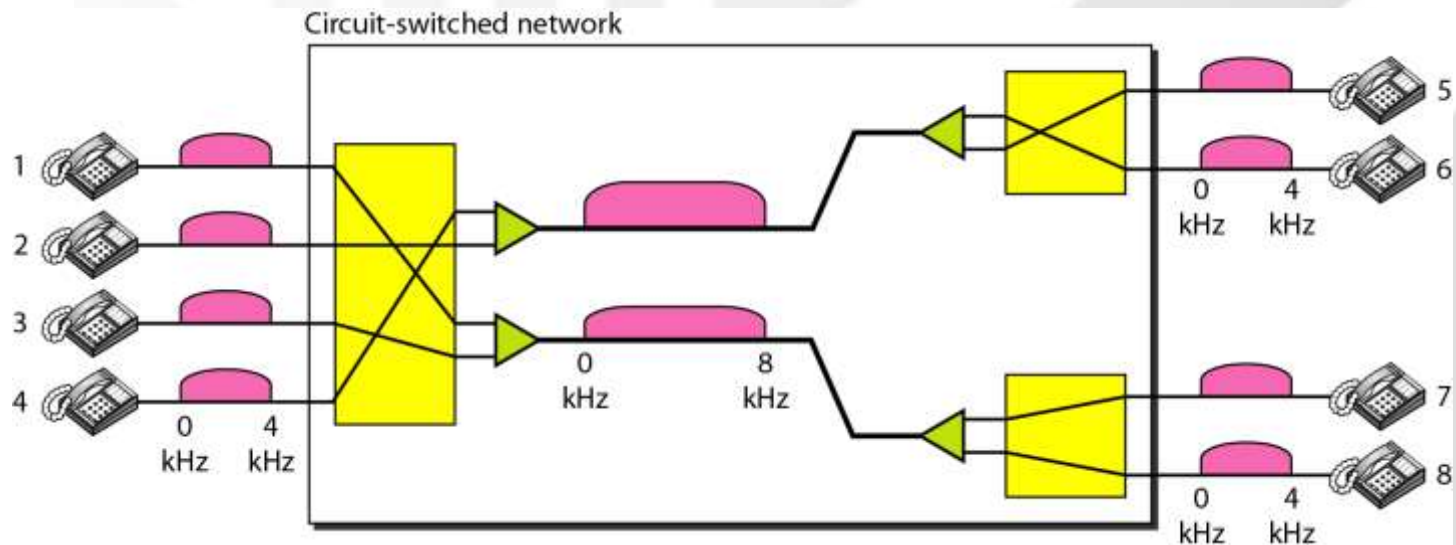
- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links.
- However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.

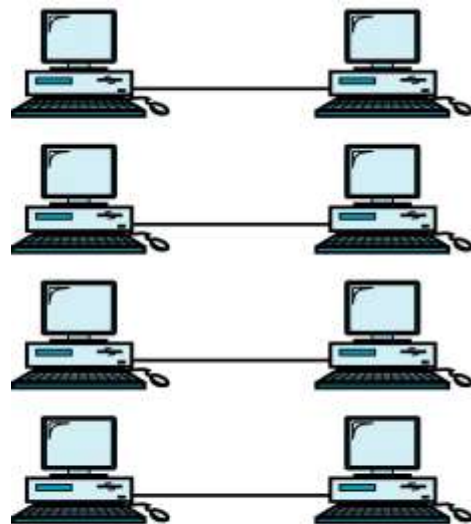


A trivial circuit-switched network

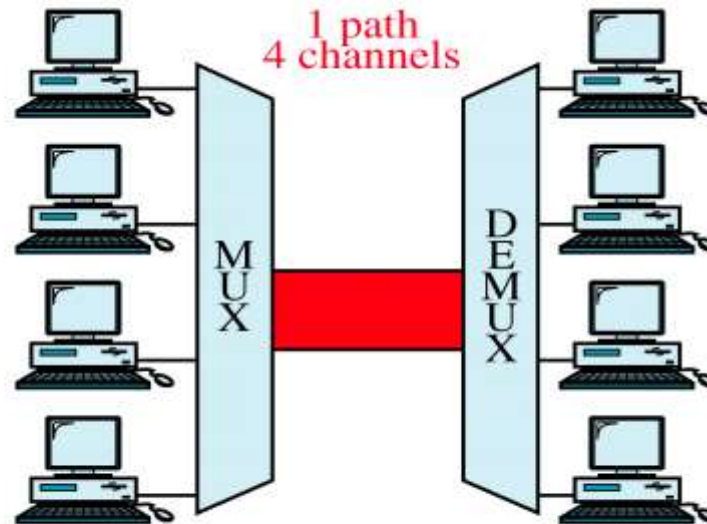
Example 1

As a trivial example, let us use a circuit-switched network to connect *eight telephones in a small area*. Communication is through *4-kHz voice channels*. We assume that each link uses FDM to connect a maximum of *two voice channels*. The bandwidth of each link is then 8 kHz. Figure 8.4 shows the situation. Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6. Of course the situation may change when new connections are made. The switch controls the connections.





a. No multiplexing



b. Multiplexing

WCB/McGraw-Hill

1. Multiplexing

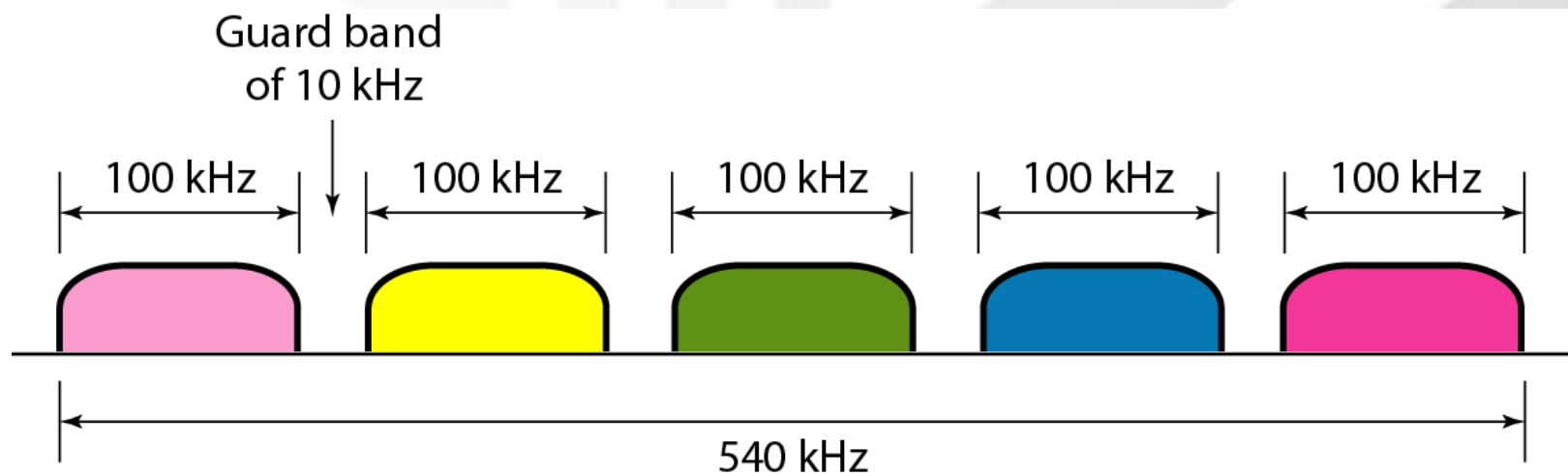
- Multiplexing is a set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
- Today there are many technologies can accommodate high bandwidth, i.e optical fiber and satellite microwave.
- Each has a bandwidth far in excess of that needed for the average transmission signal.
- If the bandwidth of a link is greater than the bandwidth needs of the devices connected to it, the bandwidth is wasted.
- Bandwidth is the precious thing in data communication.

- In multiplexed system, n lines share the bandwidth of one link.
- The sender is a **multiplexer (MUX)**.
- The receiver is a **demultiplexer (DEMUX)**
- **Link** refer to the **physical path**.
- **Channel** refer to the portion of a link that carries a transmission between a given pair of lines.
- One link can have many (n) channels.
- There are 3 basic multiplexing techniques,
 - Frequency Division Multiplexing (Analog)
 - Wavelength Division Multiplexing (Analog)
 - Time Division Multiplexing (Digital)

2. Frequency- Division Multiplexing

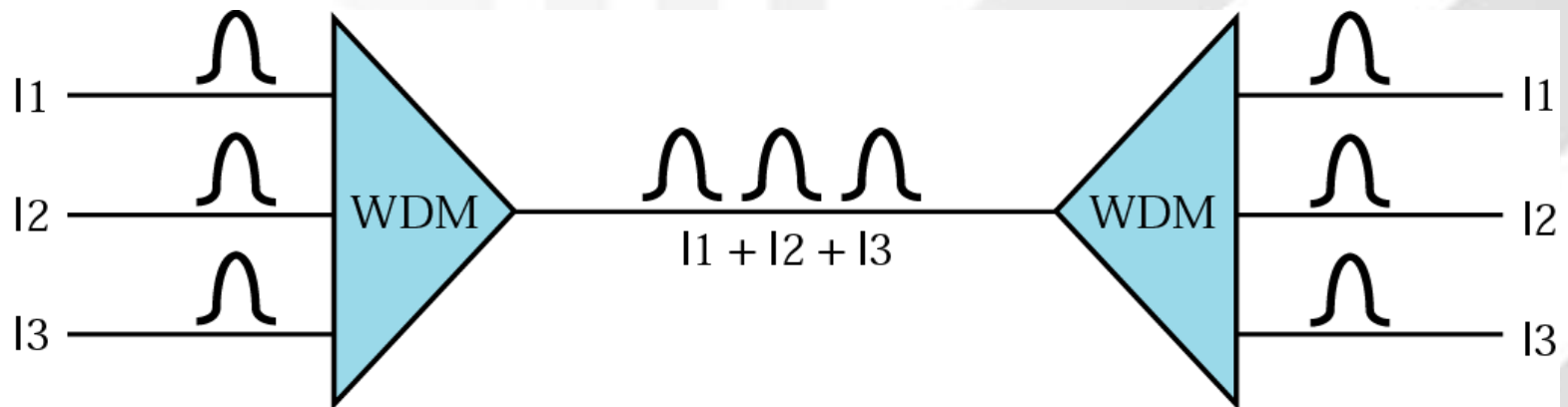
- **FDM is analog technique can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidth of the signals to be transmitted.**
- **Channels can be separated by strips of unused bandwidth – guard bands.**
- **Guard bands – to prevent signals from overlapping**

Example 2



3. Wavelength-Division Multiplexing

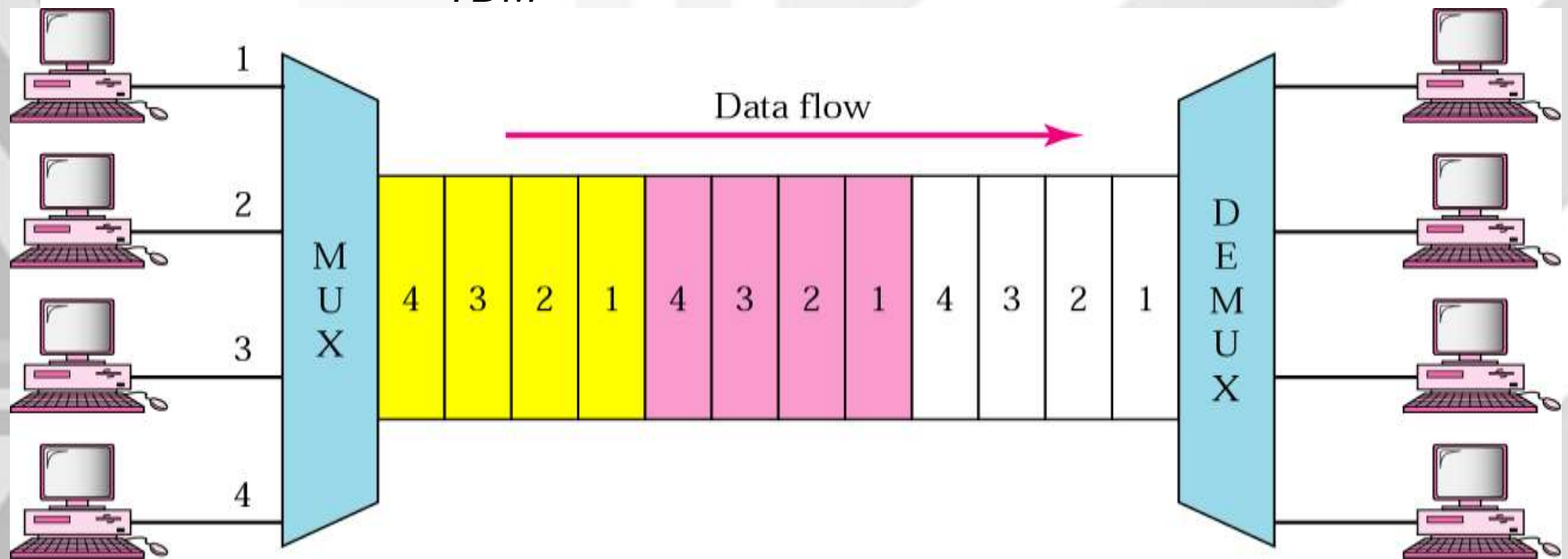
- Conceptually same as FDM, except that the multiplexing and demultiplexing involve light signals transmitted through fiber optics channels.
- The idea is same: "Combining different signals of different frequencies" the difference is that the frequencies are very high.
- Very narrow bands of light from sources are combined to make a wider band of light.
- How to combine multiple light sources into one single light at multiplexer?? → Use prism.
- Prism bends a beam of light based on the angle of incidence and the frequency.



4. Time-Division Multiplexing (TDM)

- a digital process that can be applied when the data rate capacity of **transmission medium is greater than the data rate required by the sending and receiving devices.**
- Multiple transmissions can occupy a single link by subdividing them and interleaving the portions.
- The link is shown sectioned by time rather than frequency.
- Implemented in two ways: synchronous TDM and asynchronous TDM.

TDM



Synchronous TDM

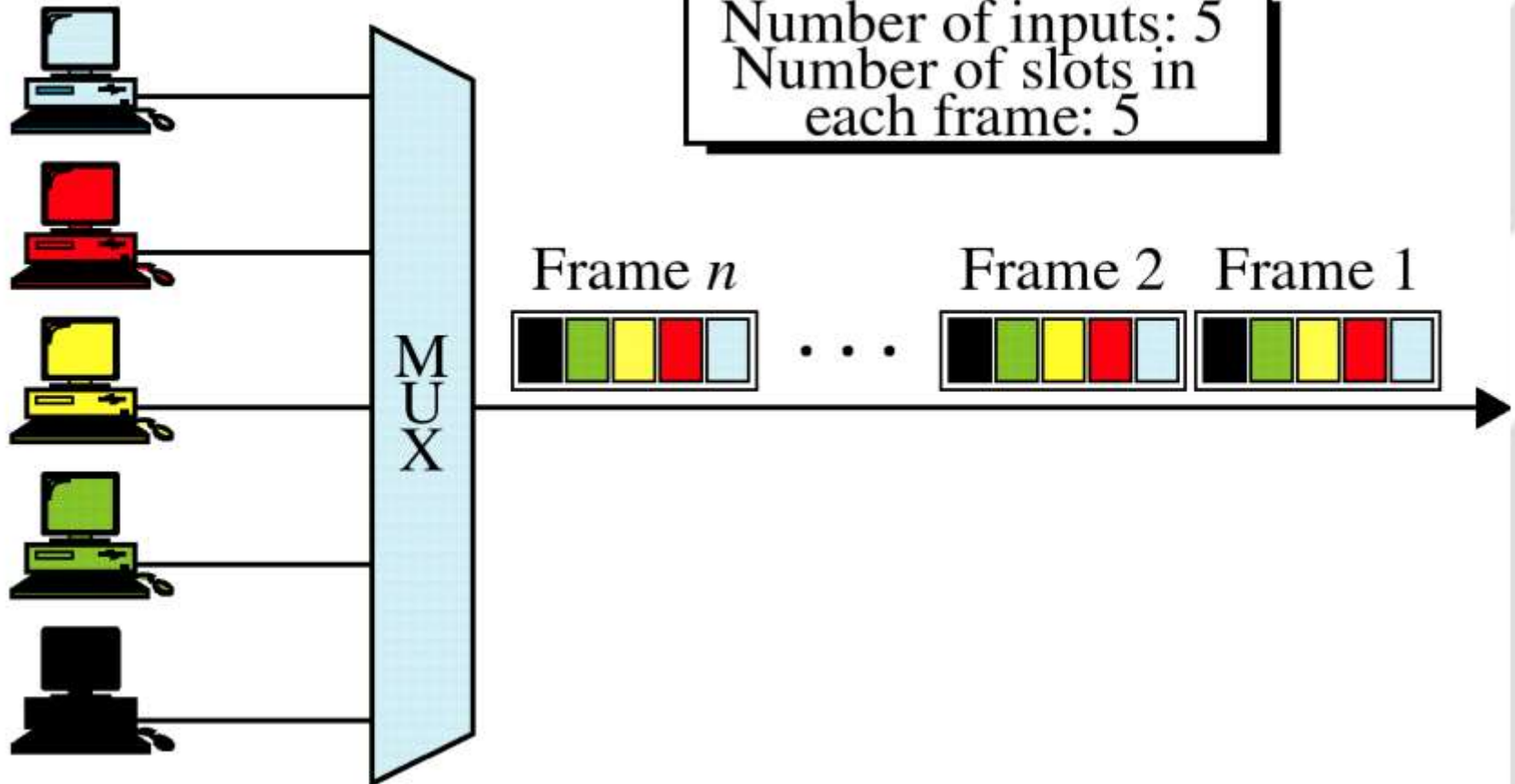
- Synchronous means that the multiplexer allocates exactly the same time slot to each device at all times, whether or not device has anything to transmit.

- Frames:

- Time slots are grouped into frames.
- A frame consists of one complete cycle of time slots, including one or more slots dedicated to each sending device
- In a system with n input lines, each frame has at least n slots.

Synchronous TDM

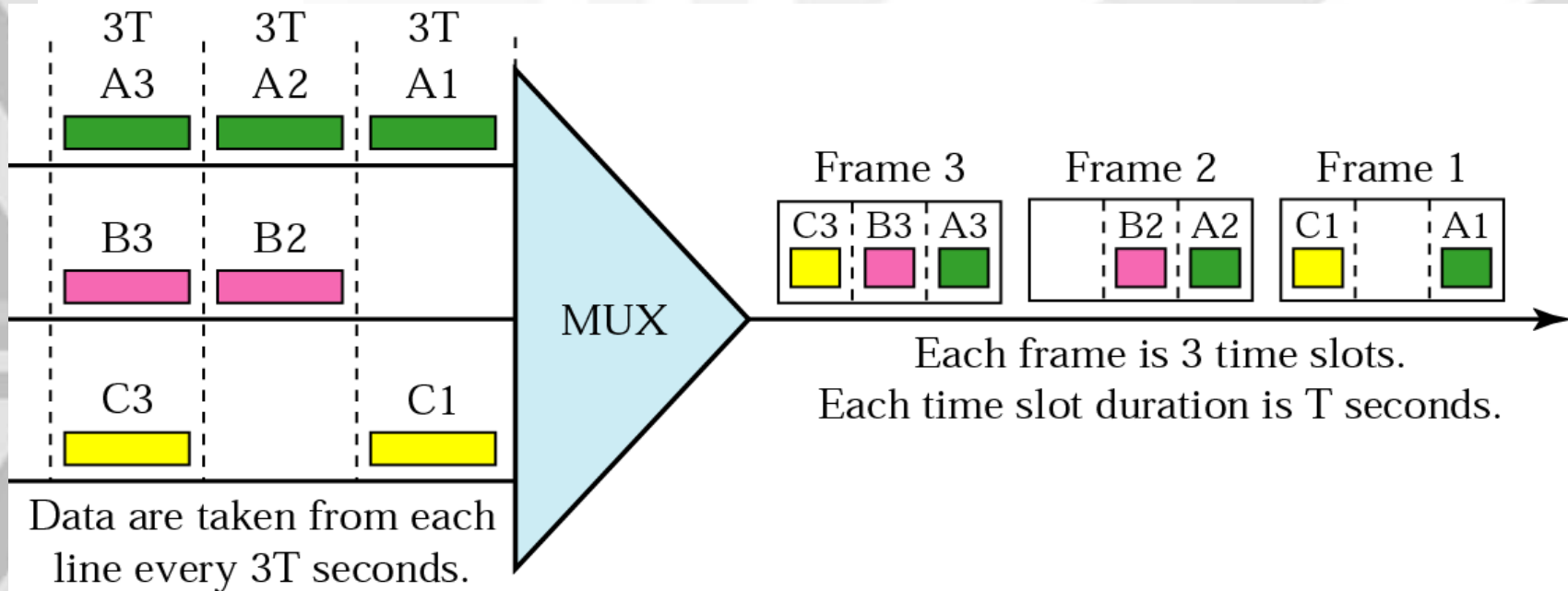
5 Inputs



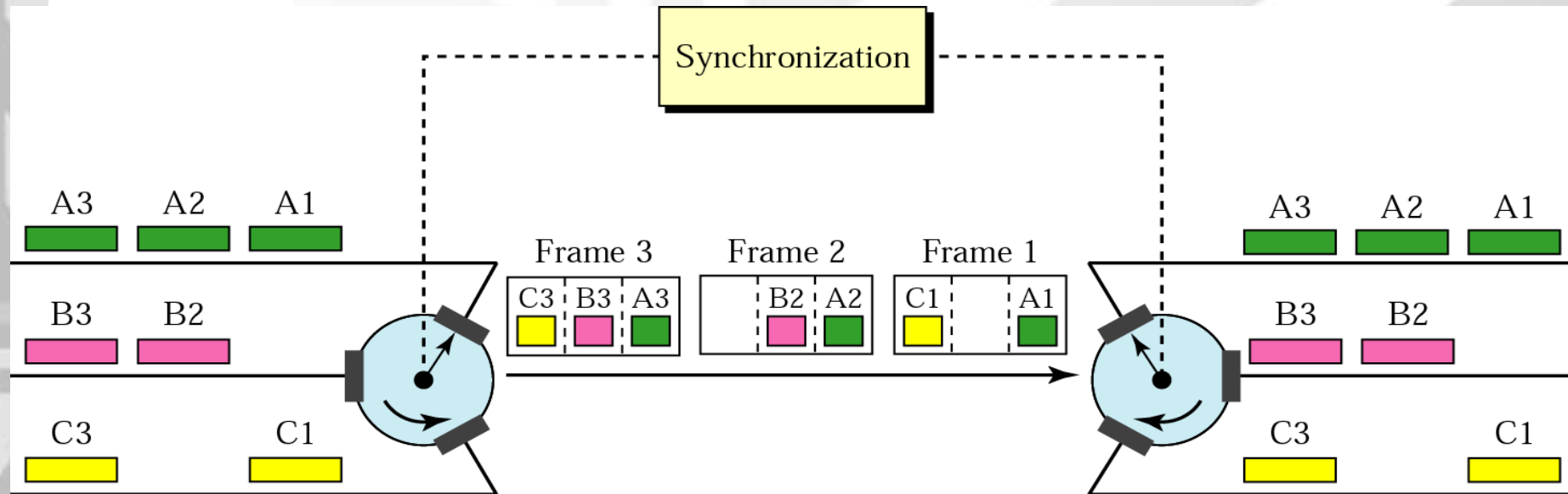
Synchronous TDM

- **Interleaving:**
 - Synchronous TDM can be compared to a very fast rotating switch.
 - The switch moves from device to device at a constant rate and in a fixed order - interleaving.
 - Interleaving can be done by bit, by byte, or by any other data unit.
- **Framing bits:**
 - The order of receipt tells the demultiplexer where to direct each time slot – so, no addressing required.

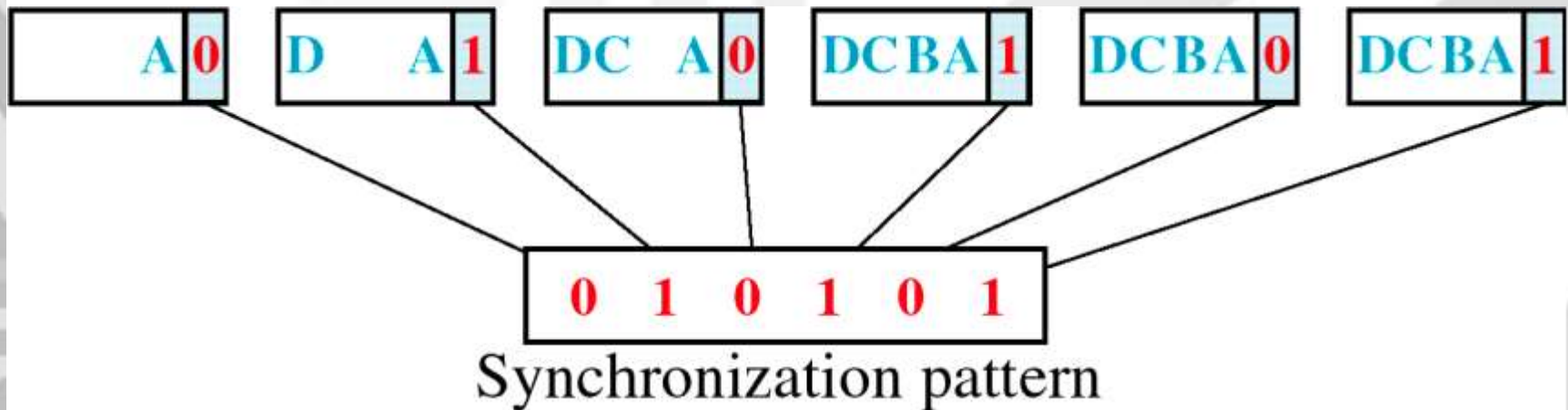
TDM frames



Interleaving



Framing Bits



Alternate 0s and 1s

Example 3

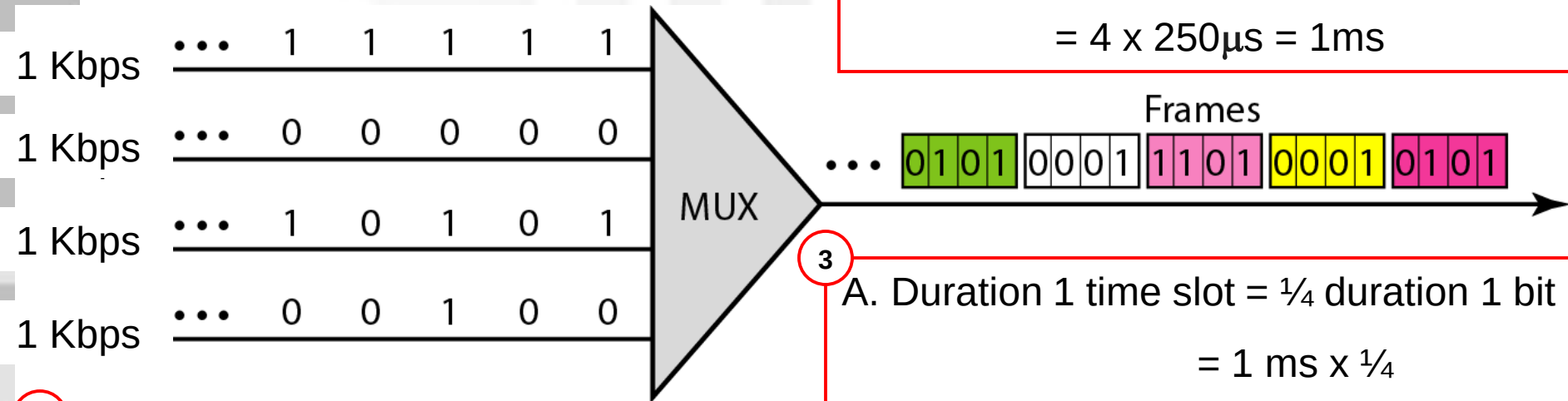
Four 1-Kbps connections are multiplexed together.

A unit is 1 bit. Find:

1. The duration of 1 bit before multiplexing $1/1 \text{ Kbps} = 0.001 \text{ s (1 ms)}$
2. The transmission rate of the link is $4 \times$ rate of a connection = 4 Kbps
3. The duration of each time slot = $1\text{ms} / 4$ or $250 \mu\text{s}$
4. The duration of a frame = duration of 1 unit before multiplexing $\Rightarrow 1 \text{ ms}$

Solution for Example 3

2
Data 1 connection = 1 Kbps
4 connections = 4 x 1 Kbps = 4 Kbps



1
1 Kbps = 1000 bit ... 1 second
1 bit = 1 / 1000 = 0.001 second = 1ms

4
A. Duration 1 frame = duration 1 bit
= 1ms

B. Duration 1 frame = 4 time slots
= 4 x 250μs = 1ms

3
A. Duration 1 time slot = $\frac{1}{4}$ duration 1 bit
= 1 ms x $\frac{1}{4}$
= 0.25 ms = 250μs

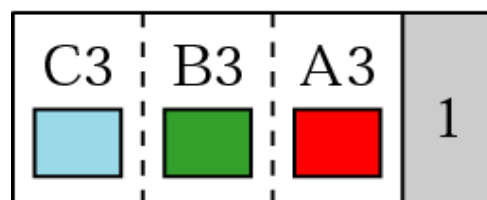
B. Duration 1 time slot = 1 / data rate
= 1 / 4 Kbps
= 0.25 ms = 250μs

Framing bits

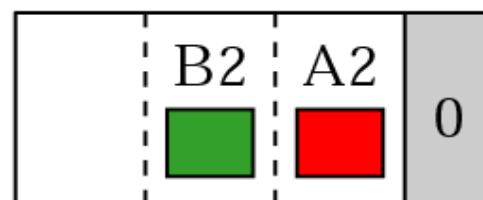
Synchronization pattern

1 0 1

Frame 3



Frame 2



Frame 1



Example 4 - optional

We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame;

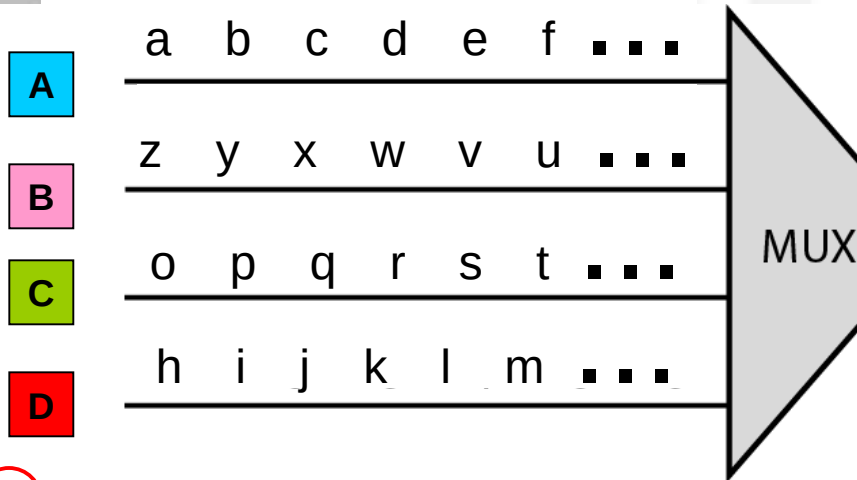
1. **The data rate of each source :** $8 \text{ bits} \times 250 = 2000\text{bps}$
2. **The duration of a character :** $1/250 \text{ s, or } 0.004\text{s} = 4\text{ms}$
3. **The frame rate :** 250 frames per second
4. **The duration of each frame :** $1 \text{ s}/250 = 4 \text{ ms}$
5. **Frame size :** $4 \text{ time slots} \times 8 \text{ bits} + 1 \text{ bit} = 33 \text{ bits.}$
6. **Data rate of output link :** $250 \text{ frame} \times 33\text{bits /s} = 8250 \text{ bps}$

Solution for Example 4

1

Data rate 1 source = 250 char x 8 bits
= 2000bps

* 1 char = 8 bits



2

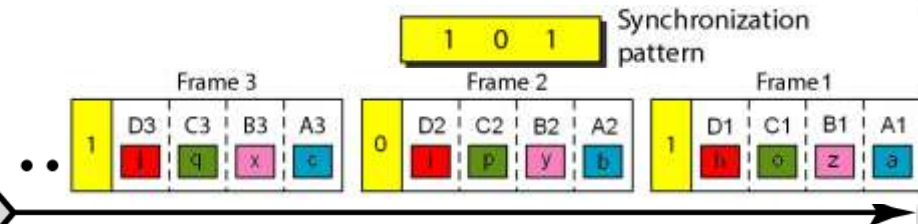
Duration of 1 char = 1 / 250
= 4 ms

3

Frame rate
= transmission
rate
= 250 frames/sec

4

Duration 1 Frame
= 1/250
= 4ms



5

1 Frame contains = 4 chars + 1 sync bit
= (4 chars x 8bit) + 1bit
= 33 bits

6

Data rate = frame rate x bits per frame
= 250 X 33 bits
= 8250 bps

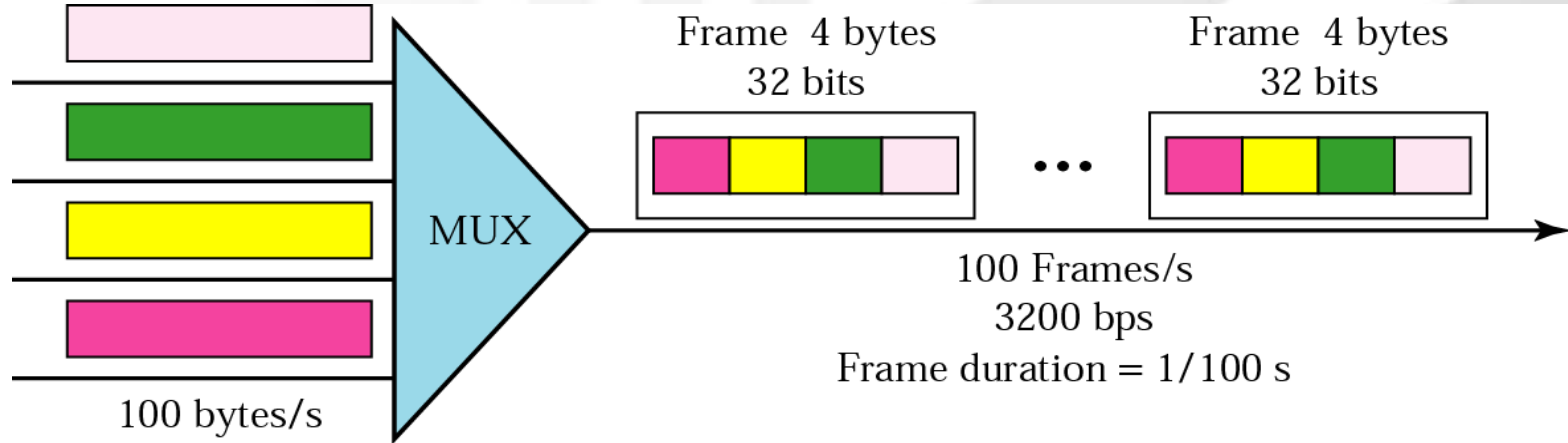
Example 5

Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Solution

The multiplexer is shown in the next figure

Example 5

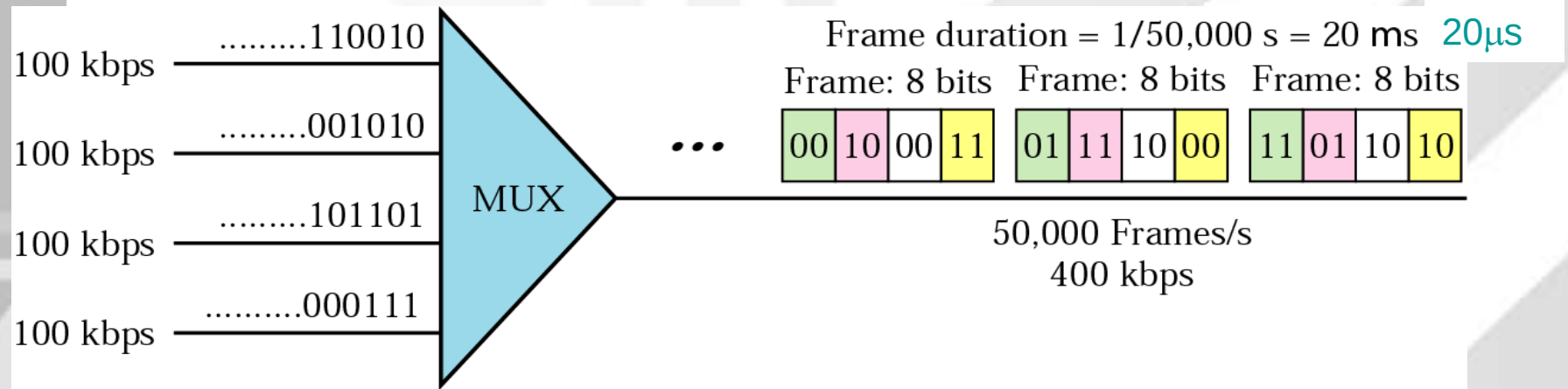


Example 6

A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

Solution

The next figure shows the output for four arbitrary inputs.



Example 7

Two channels, one with a bit rate of 100 Kbps and another with a bit rate of 200 Kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?

Solution

We can allocate one slot to the first channel and two slots to the second channel. Each frame carries 3 bits. The frame rate is 100,000 frames per second because it carries 1 bit from the first channel. The frame duration is $1/100,000$ s, or 10 ms. The bit rate is $100,000 \text{ frames/s} \times 3 \text{ bits/frame}$, or 300 Kbps.

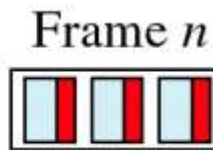
Asynchronous TDM

- Synchronous TDM does not guarantee that the full capacity of a link is used.
- Because the time slots are preassigned and fixed, whenever a connected device is not transmitting, the corresponding slot is empty and that much of the path is wasted.
- Asynchronous TDM (also called statistical TDM) is designed to avoid this type of waste.
- Asynchronous means flexible or not fixed.
- Unlike synchronous TDM, in asynchronous TDM the total speed of the input lines can be greater than the capacity of the path.

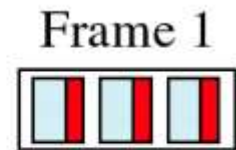
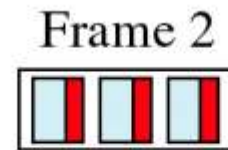
5 Inputs



Number of inputs: 5
Number of slots in each frame: 3

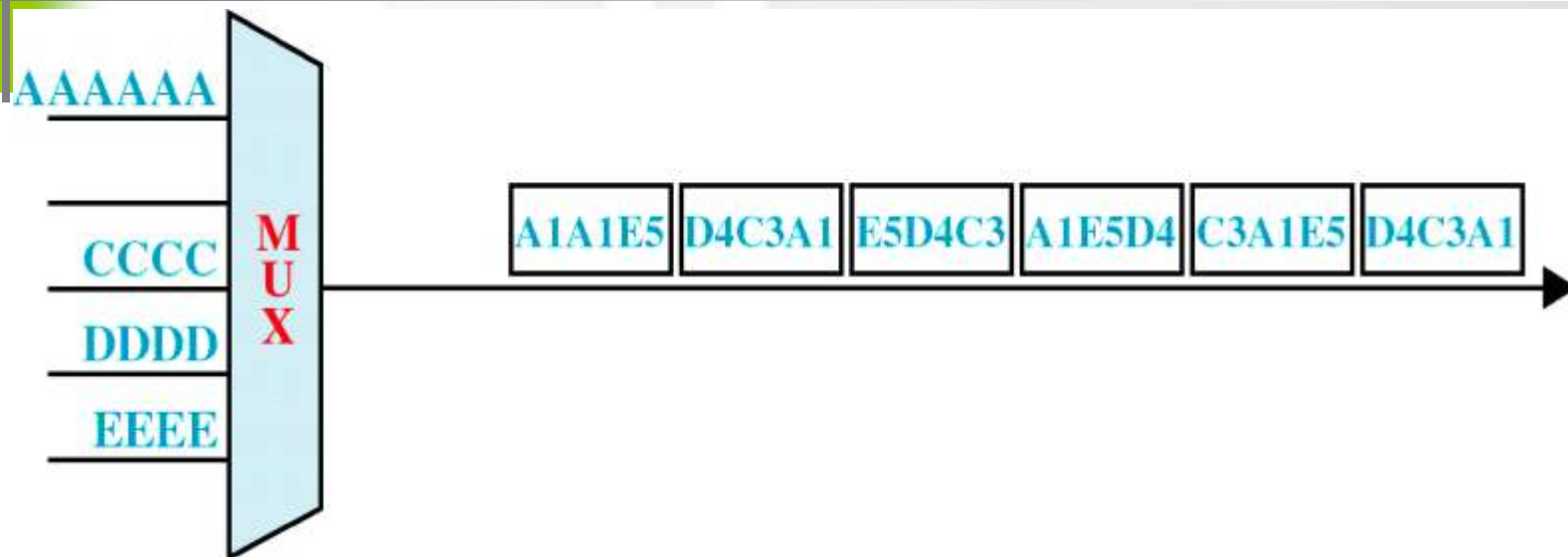


...



Asynchronous TDM

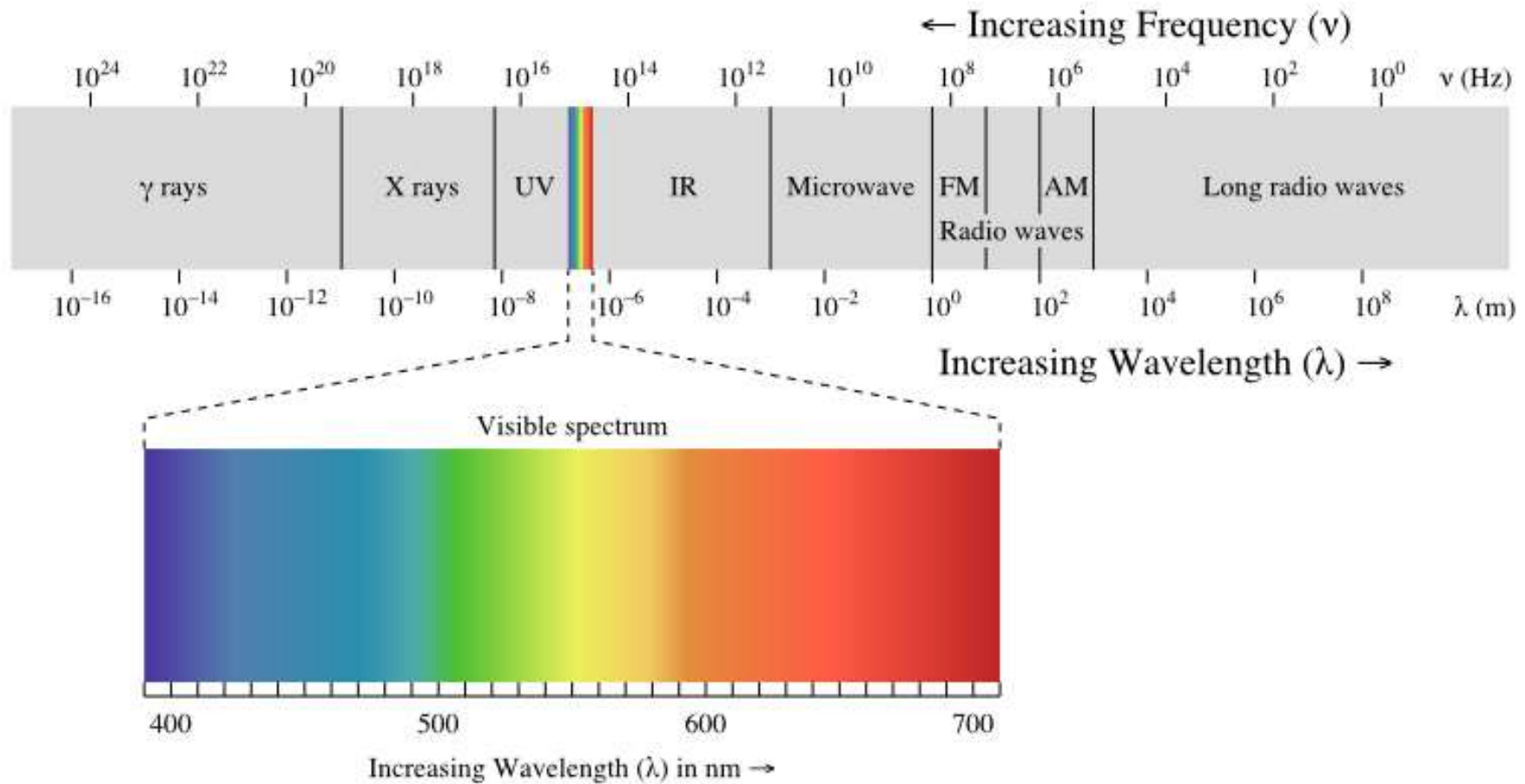
- If we have n input lines, the frame contains no more than m slots, with $m < n$.
- Given the same link, asynchronous TDM can support more devices.
- m is based on the number of input lines that are likely to be transmitted at any given time.
- Time slots allocated dynamically.

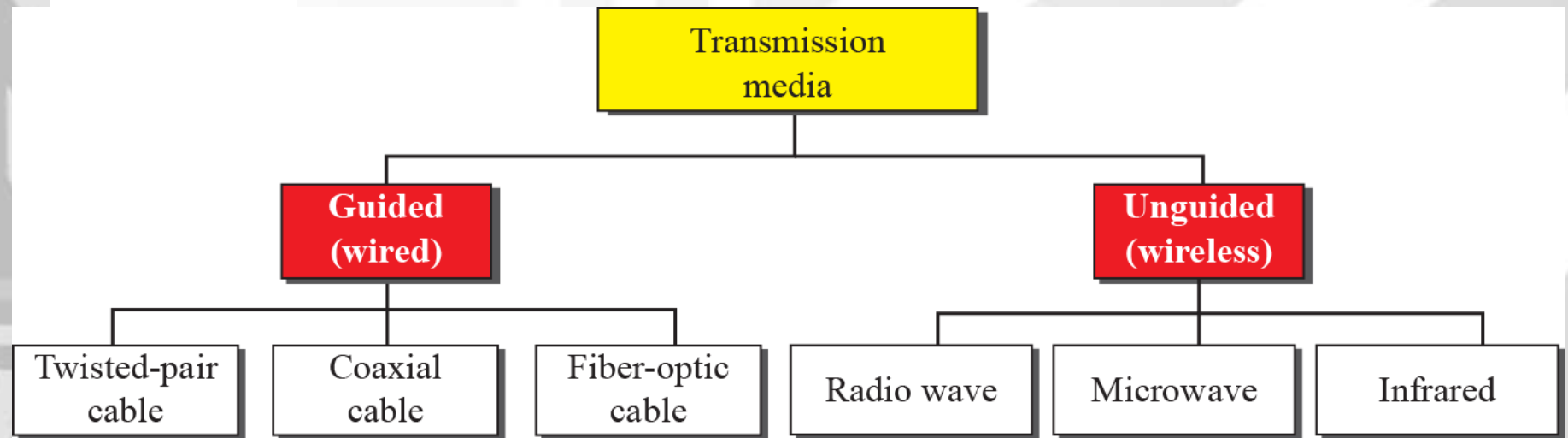


- Addressing and Overhead
 - Each time slot must carry an address telling the demultiplexer how to direct the data.
 - This address is attached by the multiplexer and discarded by the demultiplexer.

Transmission Media

Electromagnetic Signals

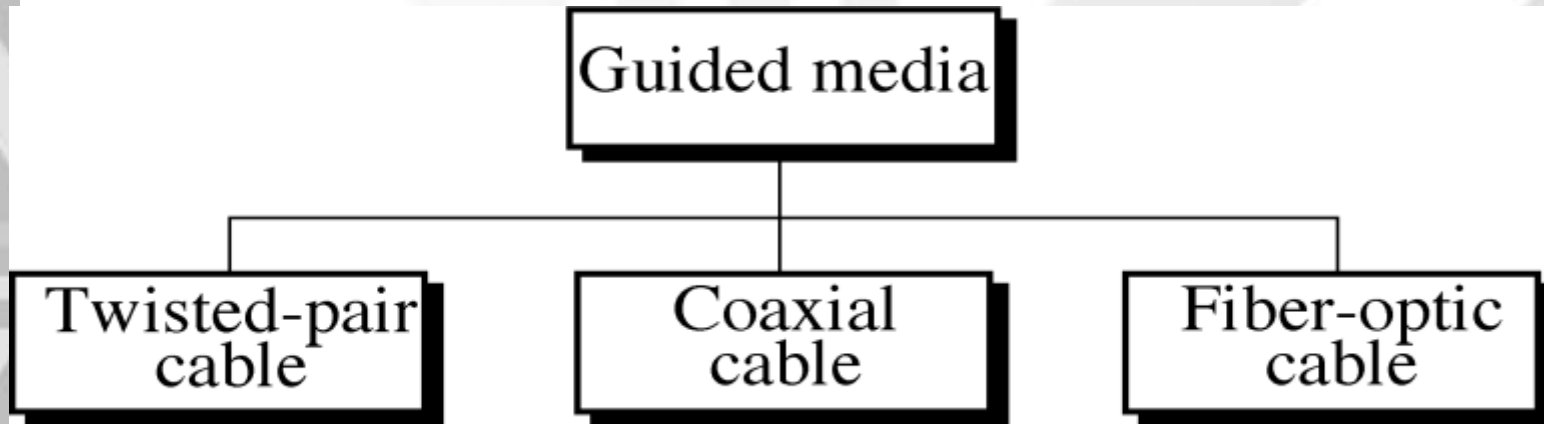




Guided Media

- **uses conduit** from one device to another.
- signal is directed and contained by the physical limits of the medium.
- Twisted pair & coaxial cable – deal with electrical current.
- Fiber optic – deal with light.

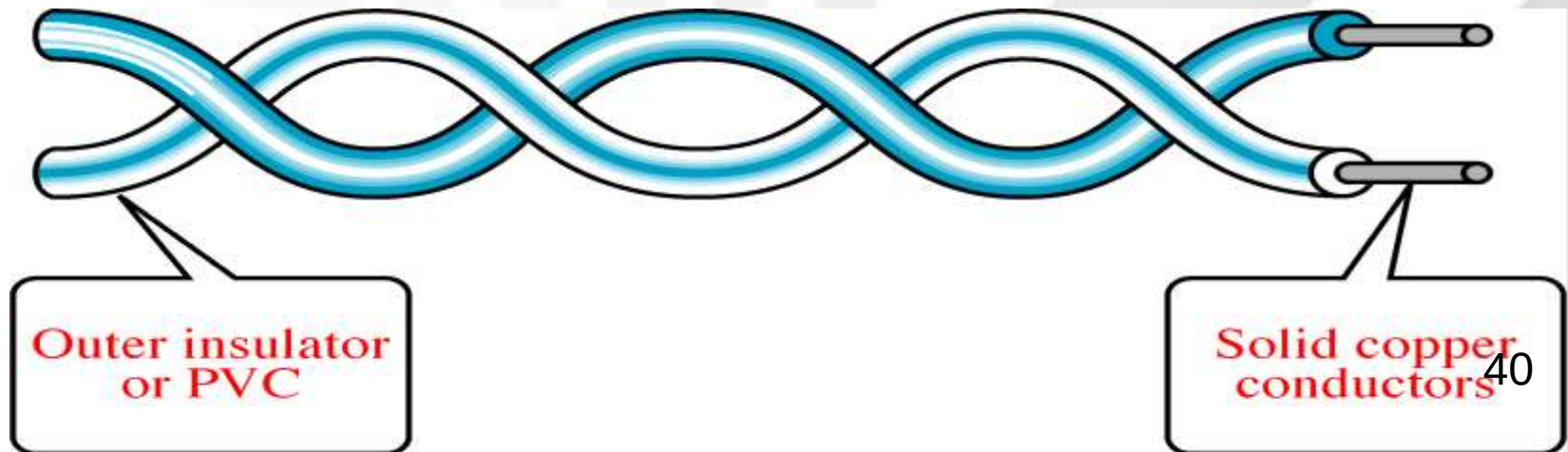
Guided Media



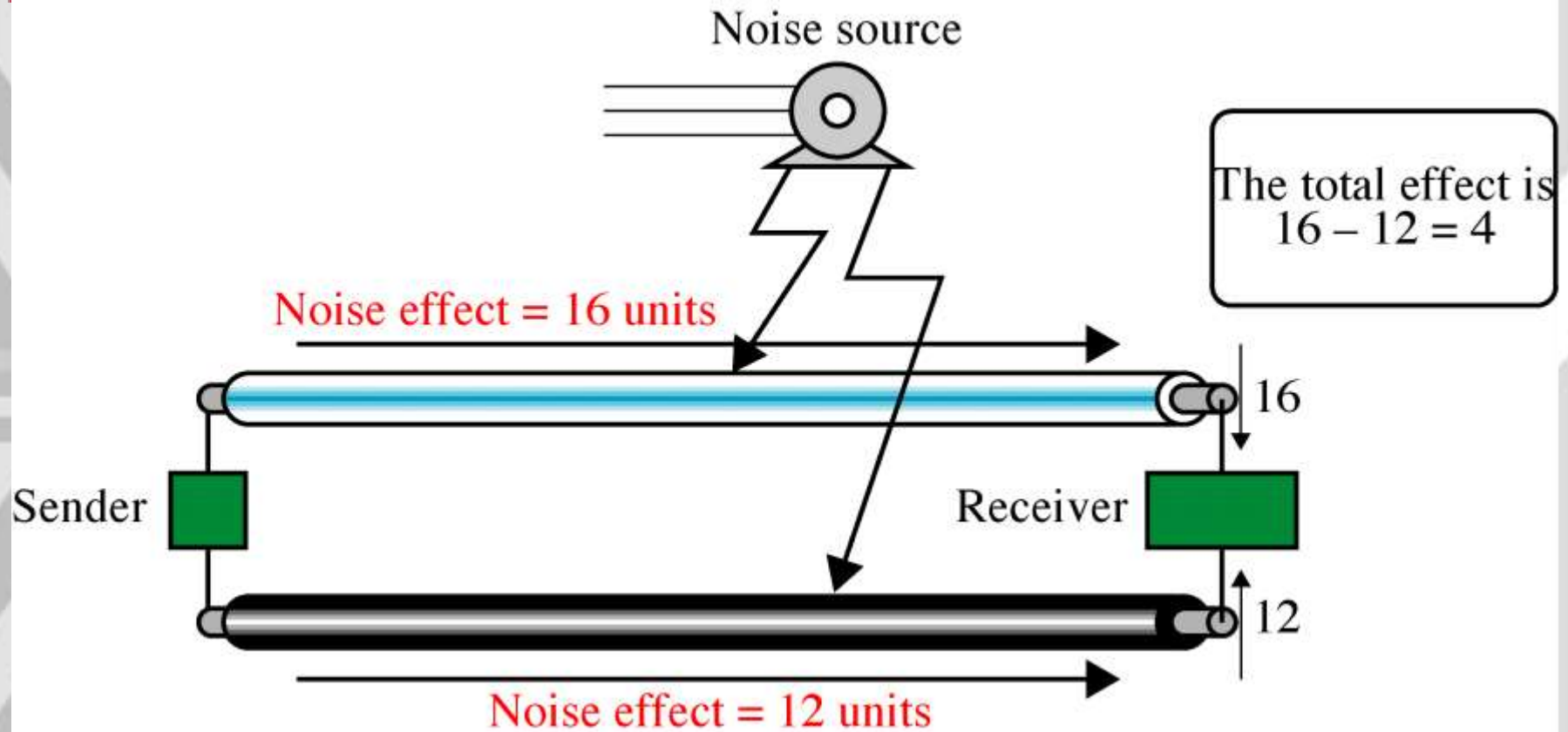
Twisted-Pair Cable

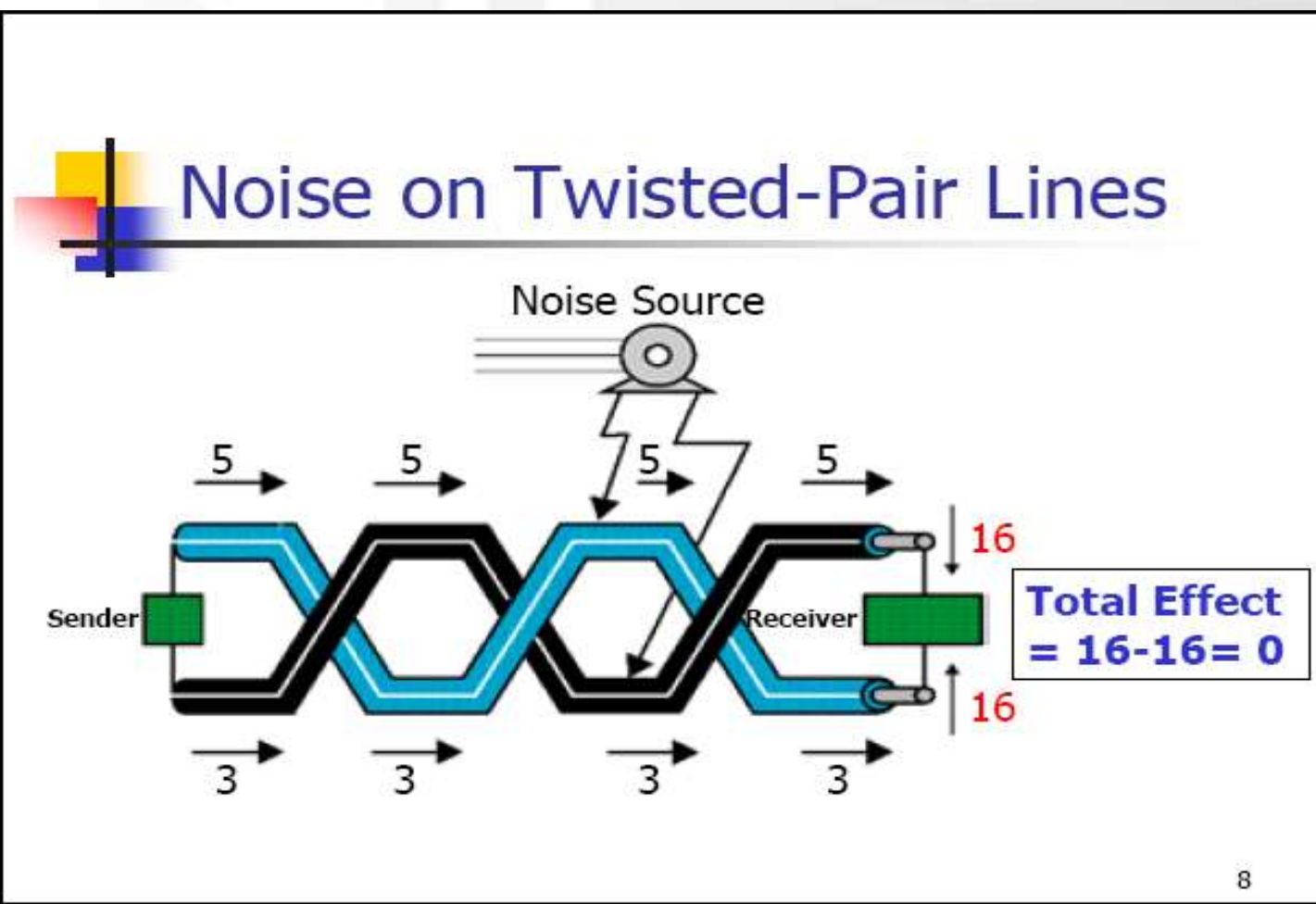
Unshielded Twisted Pair (**UTP**):

- suitable for voice and data transmission.
- consists of two conductors/copper, each with its own colored plastic insulation.
- Why use twisted instead of (**untwisted**)?



Effect of Noise on Parallel Lines





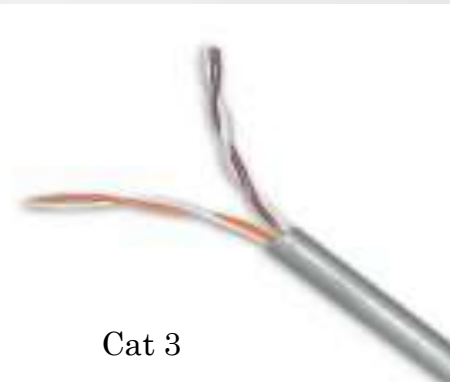
Twisted-Pair Cable

- Thin and flexible
- Small, it does not quickly fill up wiring ducts.
- Cheap

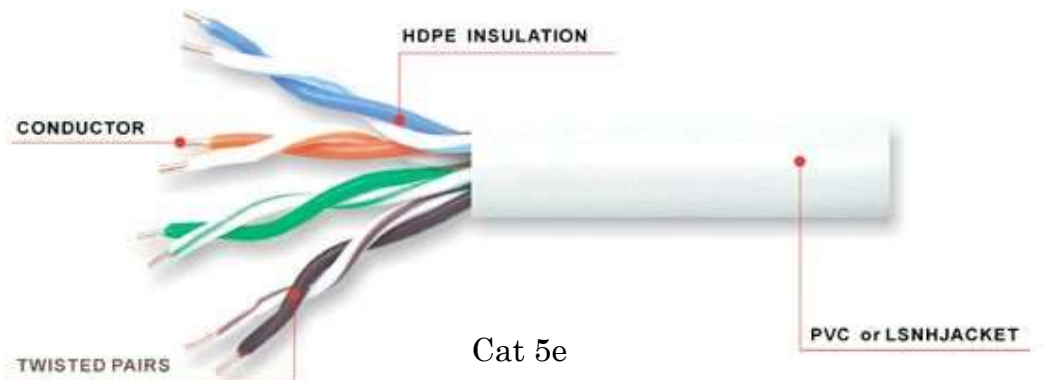
EIA has developed standards to grade UTP cables:

- Cat 3: Used for data networks using frequencies up to **16 MHz**. Historically popular for **10 Mbit/s Ethernet** networks.
- Cat 5e: Provides for up to **100 MHz**, and is frequently used for both **100 Mbit/s** and **Gigabit Ethernet** networks.
- Cat 6: Provides for up to **250 MHz**, more than double category 5 and 5e - **200 Mbit/s**
- Cat 6a: Future specification for **10 Gbit/s** applications.
- Cat 7: Designed for transmission at frequencies up to **600 MHz**.⁴³

<i>Category</i>	<i>Specification</i>	<i>Data Rate (Mbps)</i>	<i>Use</i>
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs



Cat 3



Cat 5e



Cat 6



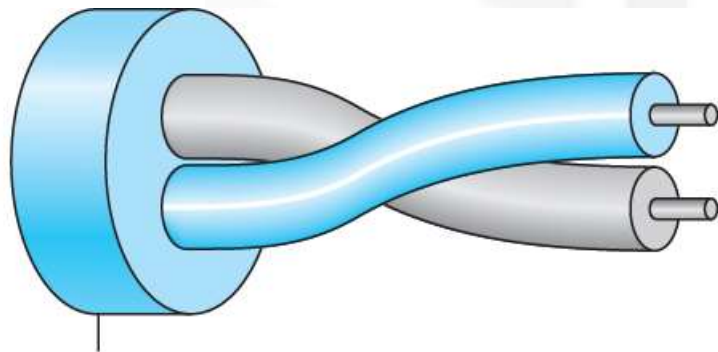
Cat 6a



Cat 7

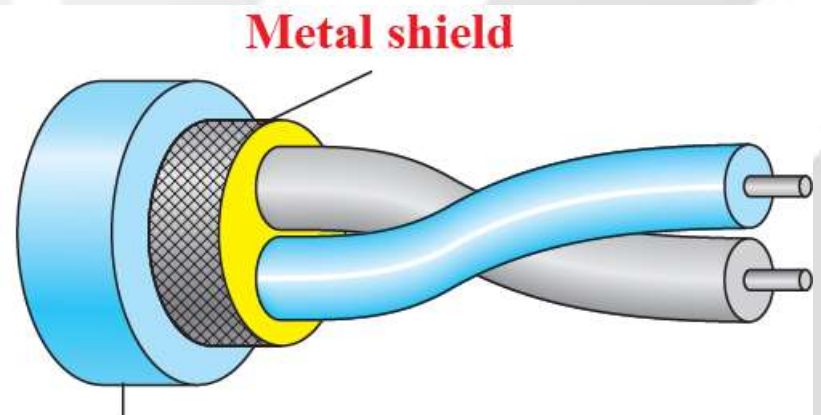
Twisted Pair Cable

- **Shielded Twisted-Pair cable (STP):**
 - Has a metal foil/braided-mesh covering that encases each pair of insulated conductors.
 - The metal casing prevents the penetration of electromagnetic noise.
 - Expensive than UTP but less susceptible to noise.
 - Eliminates crosstalk
 - It occurs when one line picks up some of the signals traveling down another line – telephone conversation.



Plastic cover

a. UTP



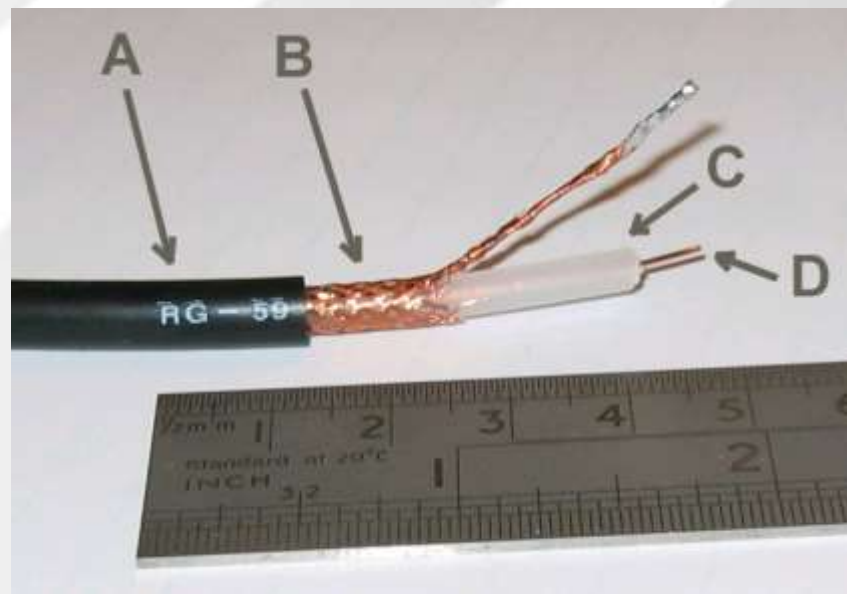
Metal shield

Plastic cover

b. STP

Coaxial Cable

- Carries signals of higher frequency ranges than twisted pair.
- Coax has a central core conductor of solid/stranded wire enclosed in an insulating sheath, and encased in an outer conductor of metal foil/braid.
- Outer conductor - serves both as a shield against noise and as second conductor.
- The outer also enclosed in an insulating sheath, and the whole cable protected by a plastic cover.



Radio-grade flexible coaxial cable.

A: outer plastic sheath

B: copper screen

C: inner dielectric insulator

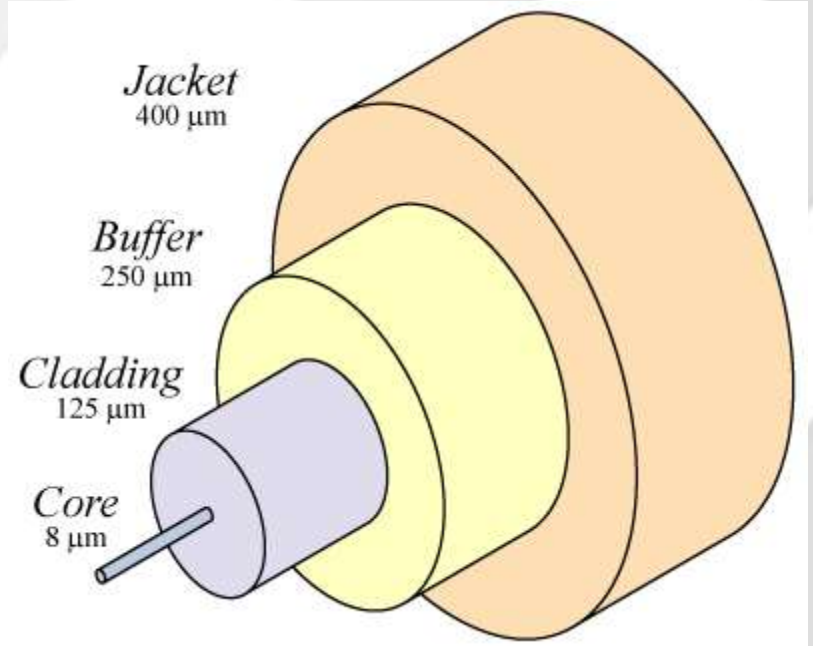
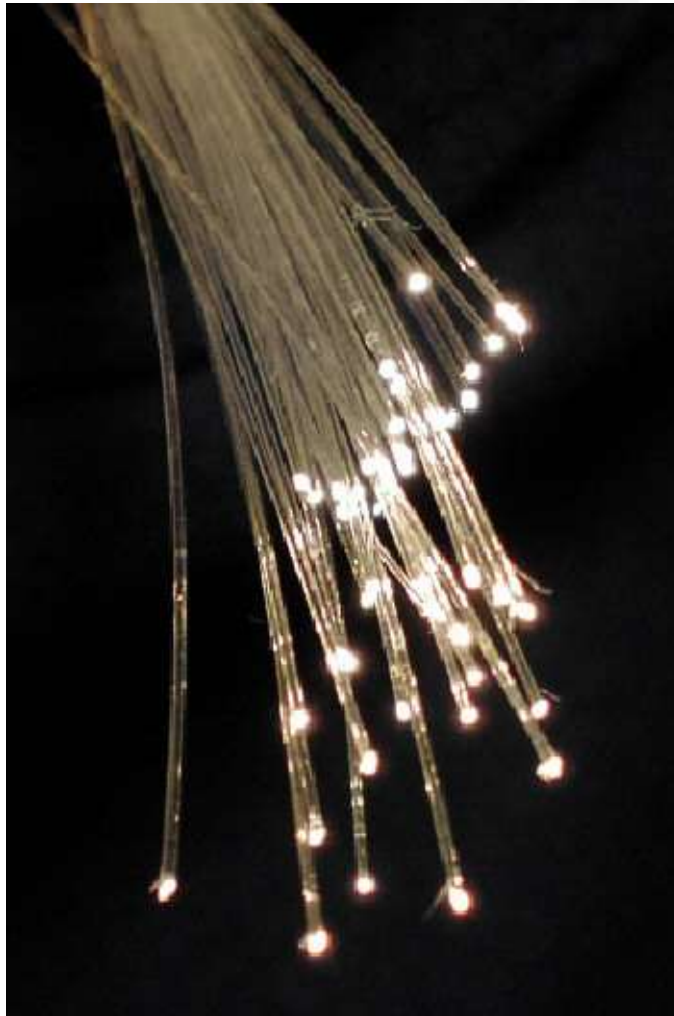
D: copper core

Coaxial Cable

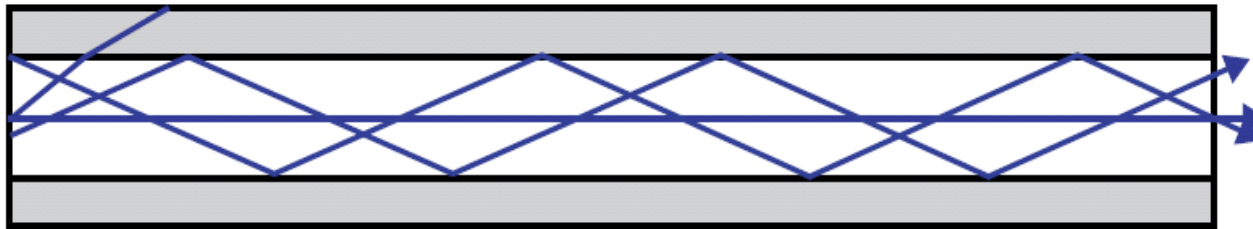
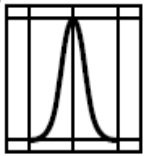
- Different coax cable designs are categorized by their radio government (RG) ratings.
- Few of the common coaxial cable standards:
 - RG-8, RG-9, RG-11. – Used in thick Ethernet
 - RG-58. Used in thin Ethernet, amateur radio
 - RG-59. Used for CCTV

Optical Fiber

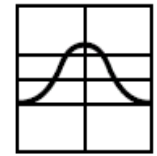
- **Optical fiber, is made of glass/plastic and transmit signals in the form of light.**
- **Light is a form of electromagnetic energy, travels at 300,000 kilometers/second in a vacuum. This speed decreases as the medium through which the light travels becomes denser.**



Input pulse

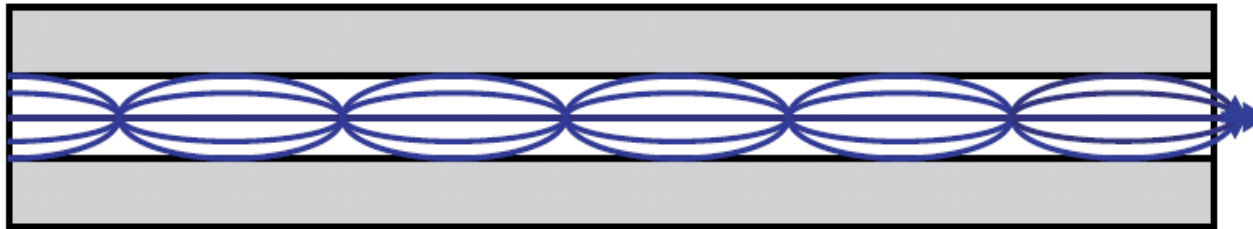
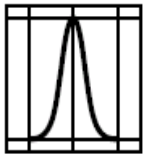


Output pulse

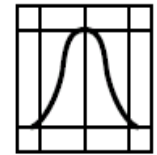


(a) Step-index multimode

Input pulse

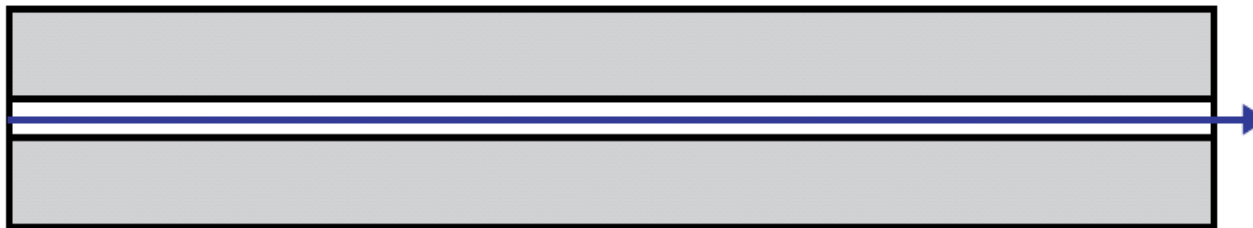
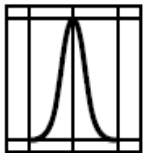


Output pulse

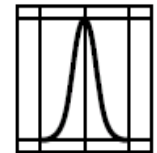


(b) Graded-index multimode

Input pulse



Output pulse



(c) Single mode

Fiber Optic

- **Advantages of Optical Fiber**
 - Noise resistance – use light rather than electricity, external light is blocked by the outer jacket
 - Less signal attenuation - transmission distance is greater than that other guided media. Without requiring regeneration
 - Higher bandwidth –compared to twisted-pair and coaxial cable
- **Disadvantages of Optical Fiber**
 - Cost – expensive manufacturing
 - Installation/maintenance -sensitive
 - Fragility – more easily broken than wire

Connectors

RJ11 to RJ45 Telephone Cable



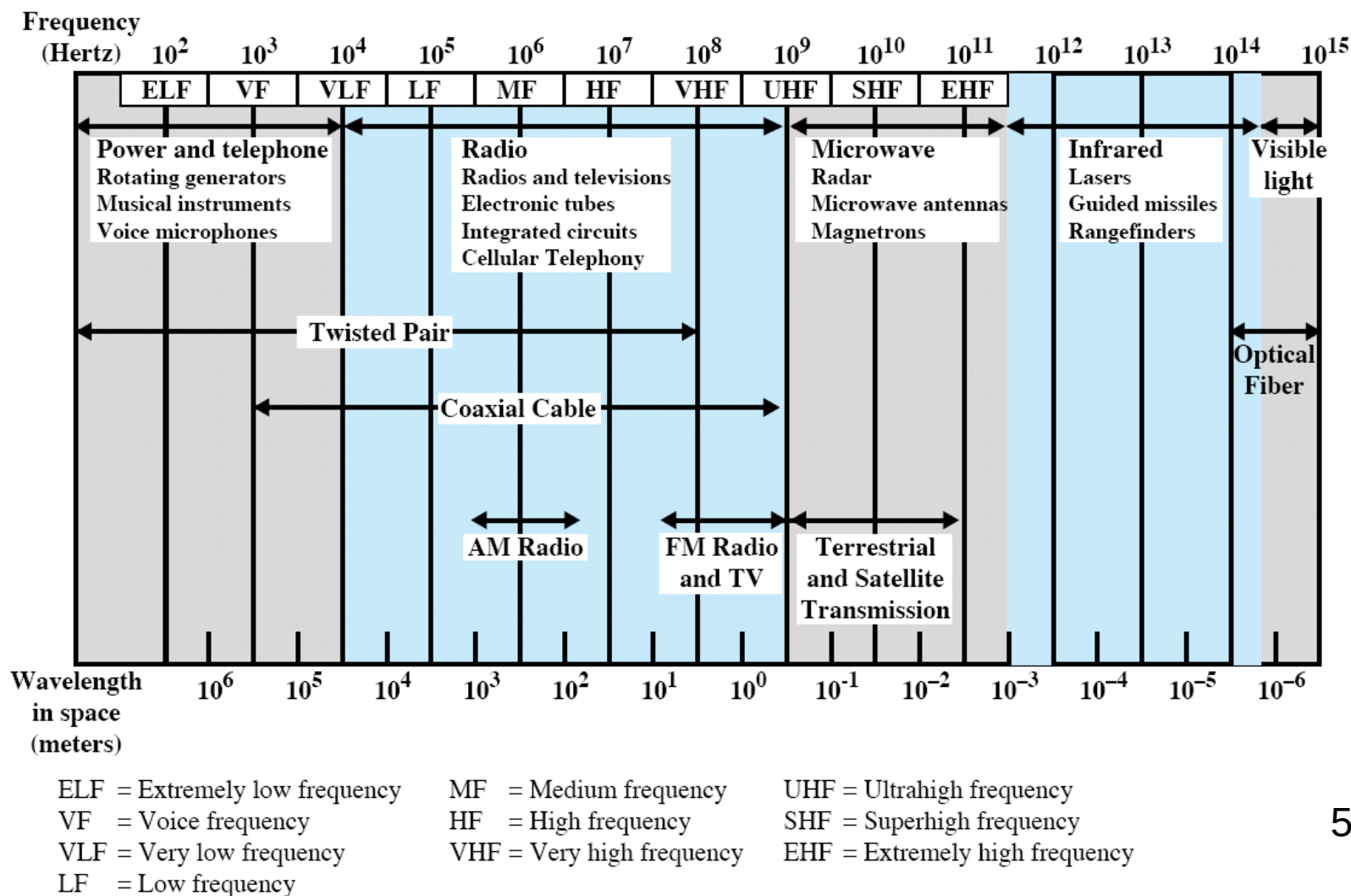
10BASE2 50 Ohm Coax Cable



Fiber Optic Cable Connectors



Electromagnetic Spectrum for Telecommunications



Unguided Media

- **Unguided media (wireless communication) transport electromagnetic waves without physical conductor**
 - **Radio waves – 3kHz-1GHz**
 - **Microwaves – 1GHz-300 GHz**

Unguided media

Omnidirectional

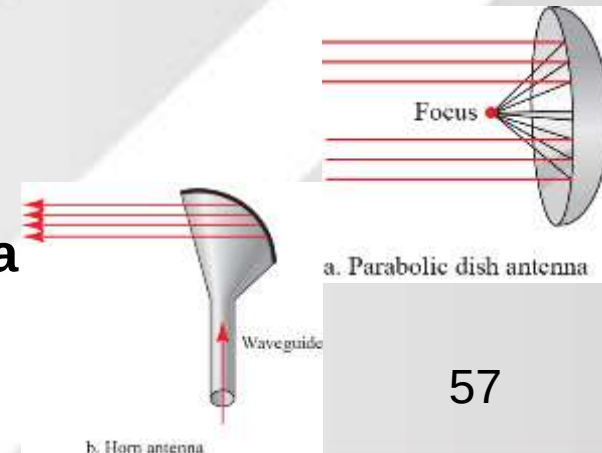
- **RADIO Waves**
- Signal radiates in all directions
- Good for broadcast
- Inexpensive antenna
- **High interference**
- Radio, normal TV



Omnidirectional antenna

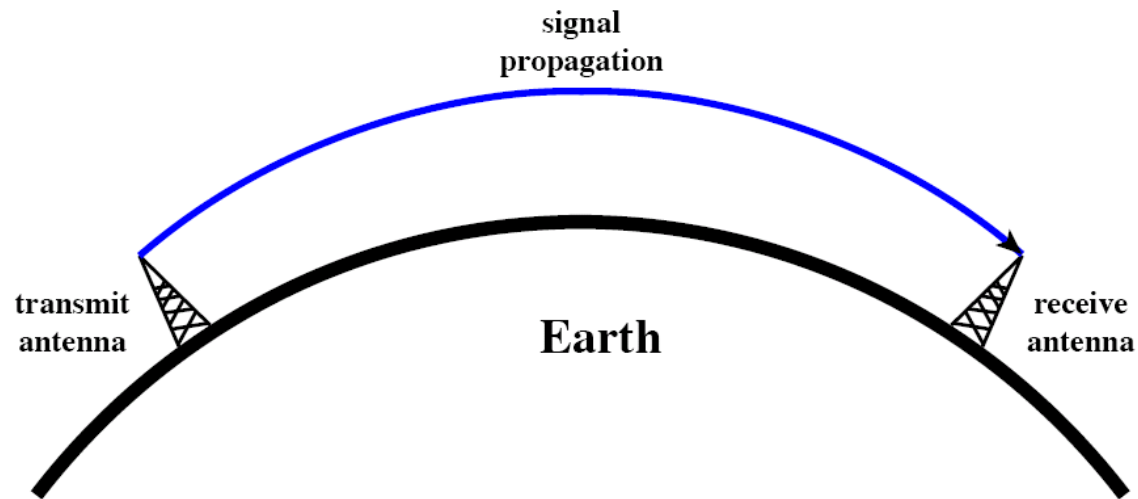
Directional

- **MICRO waves**
- Signal radiates in a single direction
- Usually requires parabolic (dish) antenna
- 2-40 GHz (microwave)
- Also works with lasers
- Cable TV (ASTRO)



Ground propagation

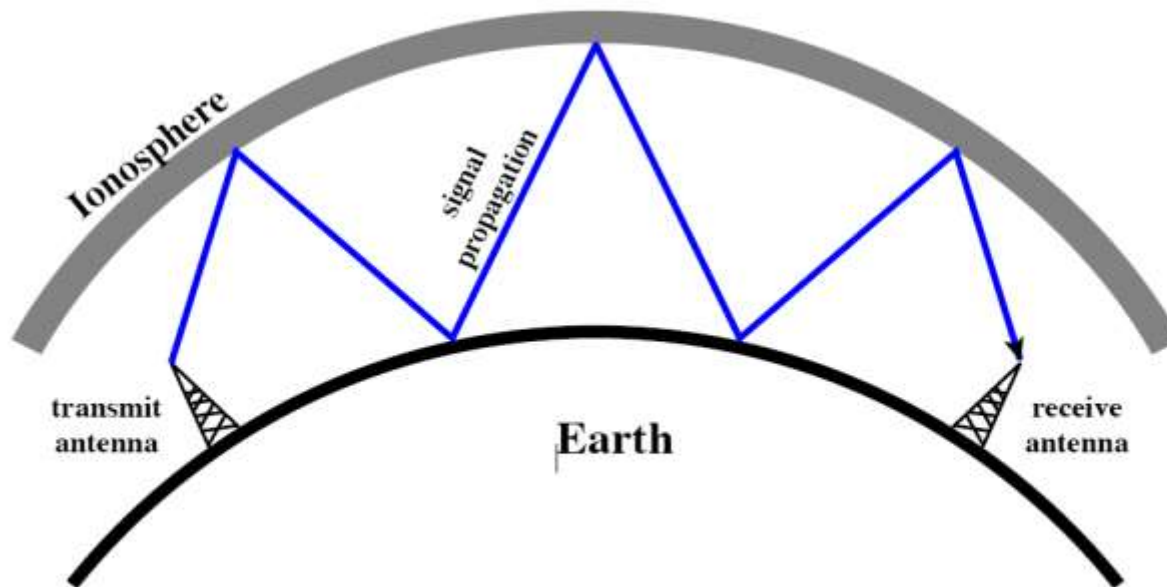
- Lowest portion of atmosphere
- Follows earth's contour



(a) Ground-wave propagation (below 2 MHz)

Sky propagation

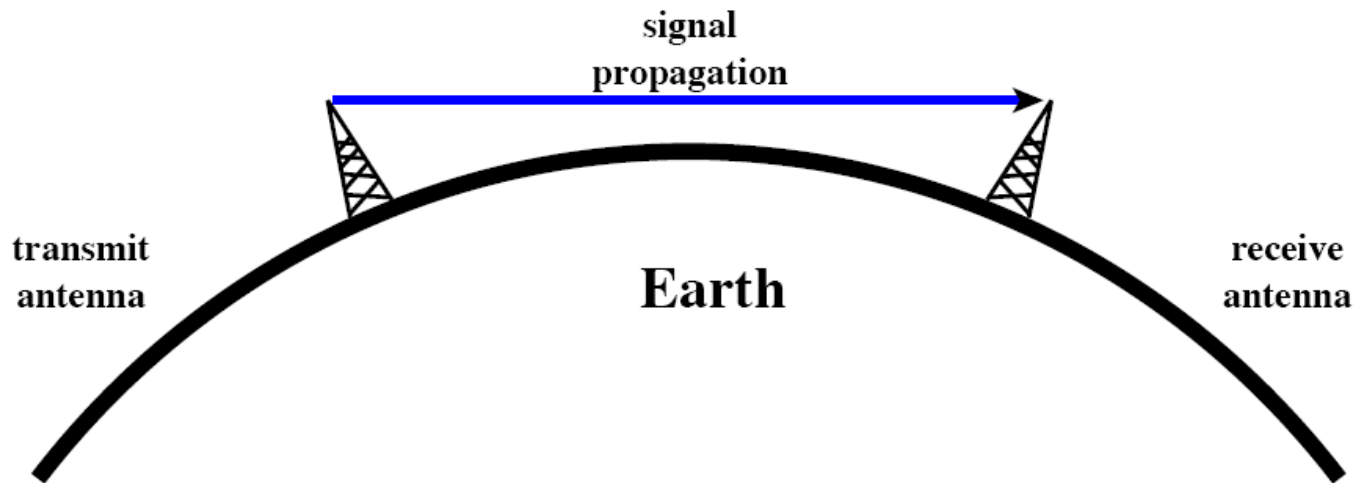
- Signal reflected at the ionosphere.



(b) Sky-wave propagation (2 to 30 MHz)

Line-of-sight propagation

- Effective straight line with antennas facing each other



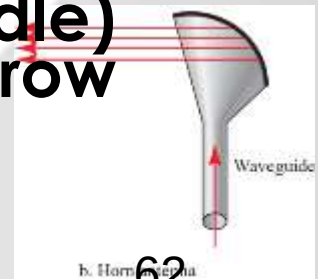
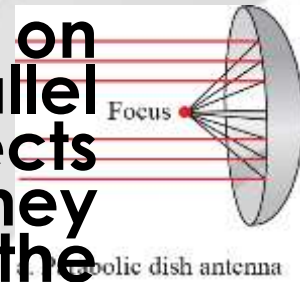
(c) Line-of-sight (LOS) propagation (above 30 MHz)

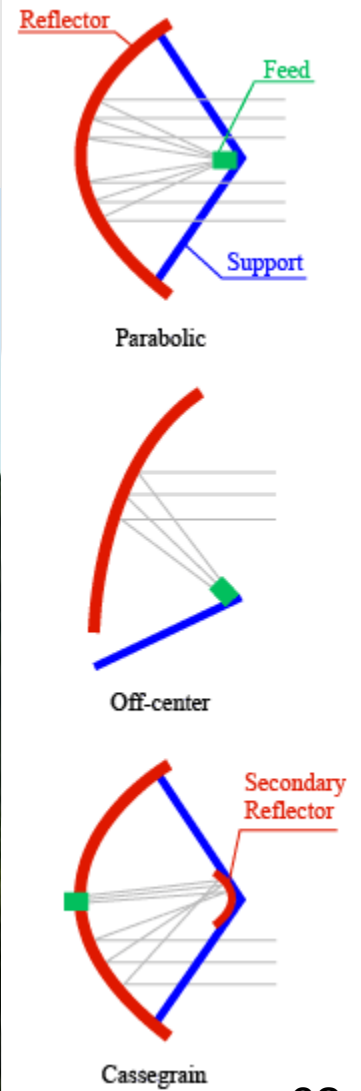
Table 4.7 Frequency Bands

Band	Frequency Range	Free-Space Wavelength Range	Propagation Characteristics	Typical Use
ELF (extremely low frequency)	30 to 300 Hz	10,000 to 1000 km	GW	Power line frequencies; used by some home control systems.
VF (voice frequency)	300 to 3000 Hz	1000 to 100 km	GW	Used by the telephone system for analog subscriber lines.
VLF (very low frequency)	3 to 30 kHz	100 to 10 km	GW; low attenuation day and night; high atmospheric noise level	Long-range navigation; submarine communication
LF (low frequency)	30 to 300 kHz	10 to 1 km	GW; slightly less reliable than VLF; absorption in daytime	Long-range navigation; marine communication radio beacons
MF (medium frequency)	300 to 3000 kHz	1,000 to 100 m	GW and night SW; attenuation low at night, high in day; atmospheric noise	Maritime radio; direction finding; AM broadcasting.
HF (high frequency)	3 to 30 MHz	100 to 10 m	SW; quality varies with time of day, season, and frequency.	Amateur radio; international broadcasting, military communication; long-distance aircraft and ship communication
VHF (very high frequency)	30 to 300 MHz	10 to 1 m	LOS; scattering because of temperature inversion; cosmic noise	VHF television; FM broadcast and two-way radio, AM aircraft communication; aircraft navigational aids
UHF (ultra high frequency)	300 to 3000 MHz	100 to 10 cm	LOS; cosmic noise	UHF television; cellular telephone; radar; microwave links; personal communications systems
SHF (super high frequency)	3 to 30 GHz	10 to 1 cm	LOS; rainfall attenuation above 10 GHz; atmospheric attenuation due to oxygen and water vapor	Satellite communication; radar; terrestrial microwave links; wireless local loop
EHF (extremely high frequency)	30 to 300 GHz	10 to 1 mm	LOS; atmospheric attenuation due to oxygen and water vapor	Experimental; wireless local loop
Infrared	300 GHz to 400 THz	1 mm to 770 nm	LOS	Infrared LANs; consumer electronic applications
Visible light	400 THz to 900 THz	770 nm to 330 nm	LOS	Optical communication

Terrestrial Microwave

- Two types of antenna are used: parabolic dish and horn
- **Parabolic** dish antenna – based on geometry of a parabola: every line parallel to the line of symmetry (line-of-sight) reflects off the curve at angles such that they intersect in a common point called the focus.
- **Horn** antenna – Outgoing transmission are broadcast up a stem (resembling a handle) and deflected outward in a series of narrow parallel beams by the curved head.





Infrared

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors.

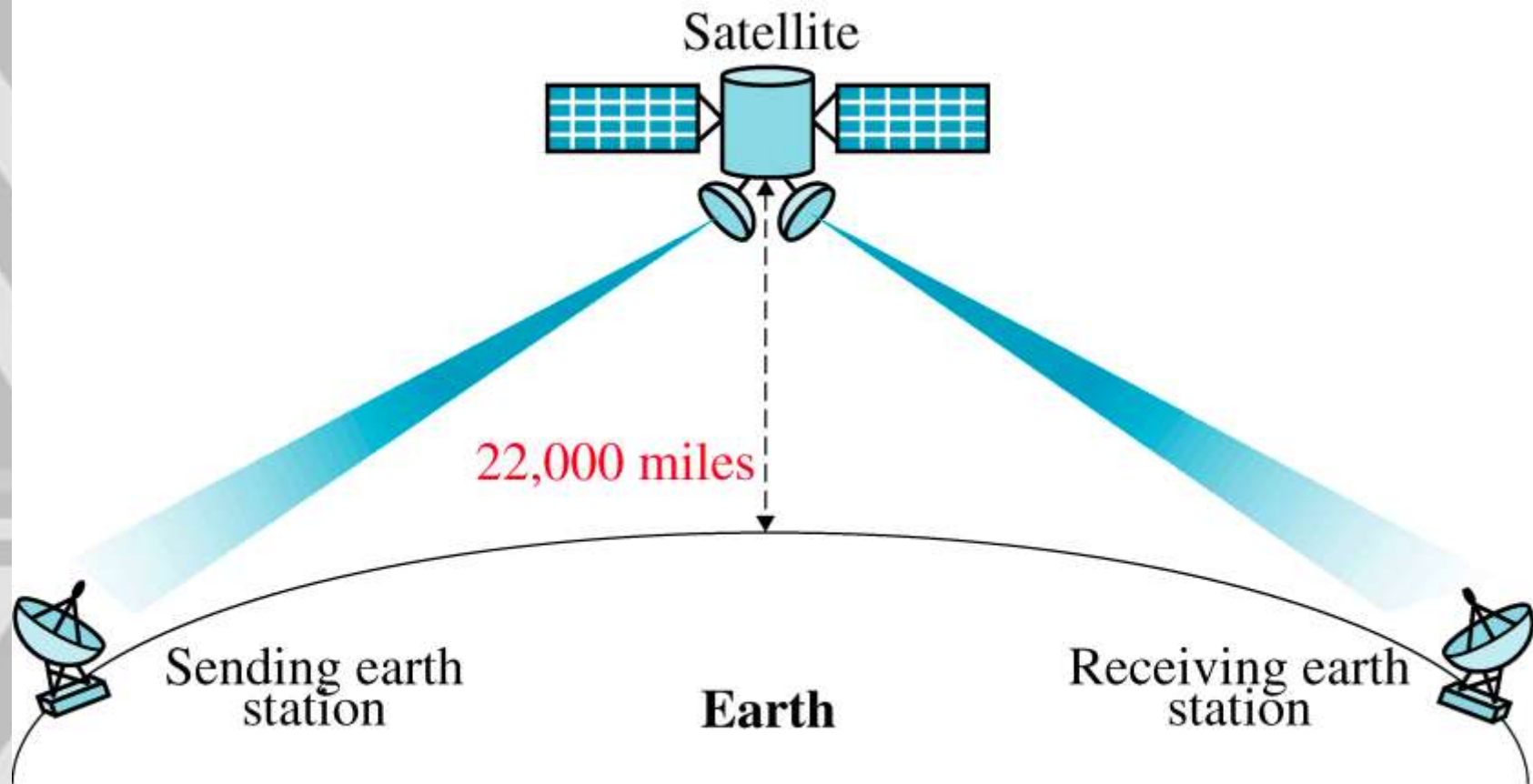
Satellite Communication

- Much like line-of-sight microwave transmission in which one of the stations is a satellite orbiting the earth.
- The principle is the same as terrestrial microwave, with satellite acting as a super tall antenna and repeater.
- Satellite microwave can provide transmission capability to and from any location on earth, no matter how remote.

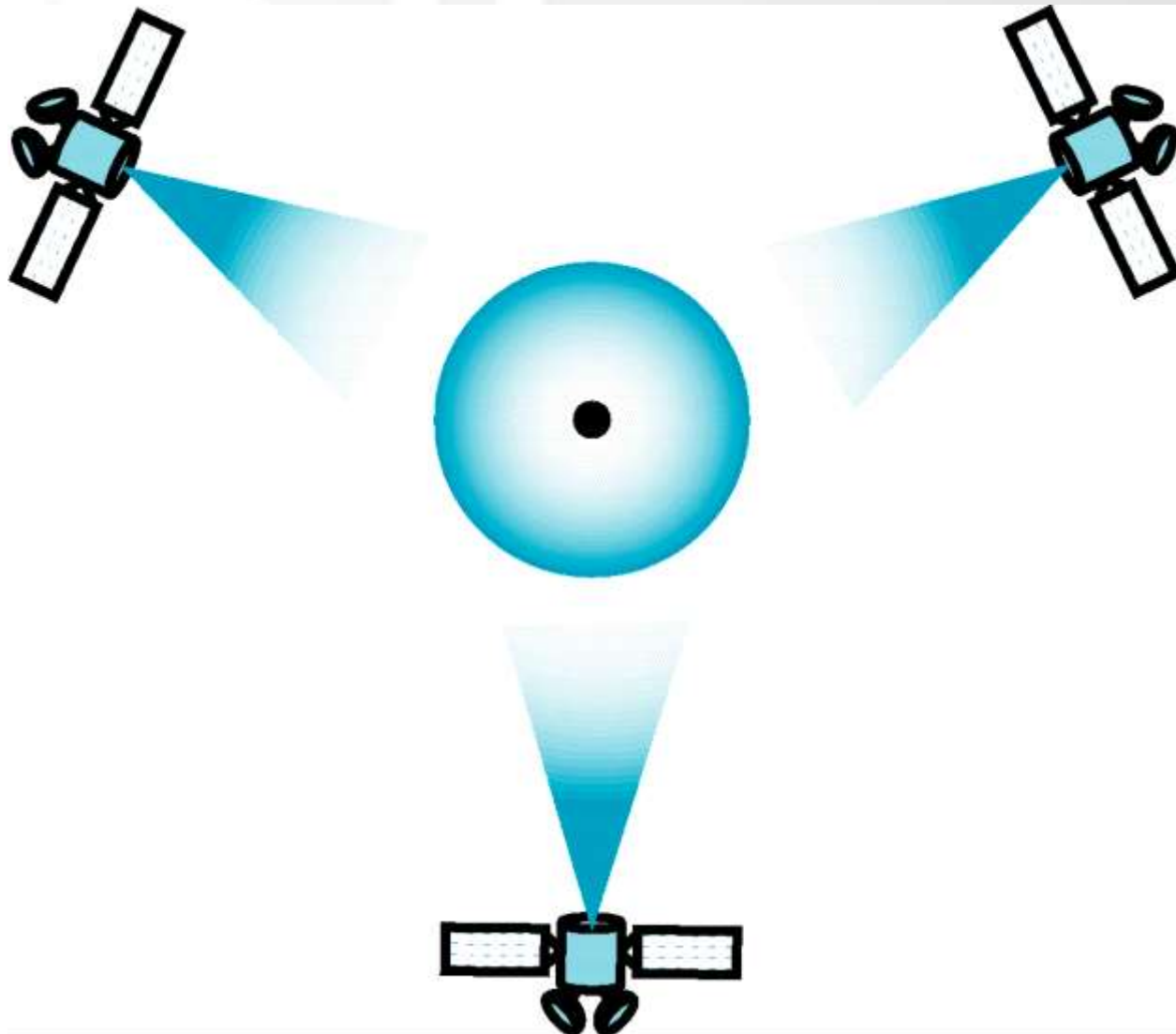
Satellite Communication

- **Geosynchronous Satellites**
 - The satellite move at the same speed as the earth to ensure constant communication.
 - Because orbital speed is based on distance from the planet, only one orbit can be geosynchronous.
 - One geosynchronous satellite cannot cover the whole earth.
 - It takes a minimum of three satellites equitant from each other in geosynchronous orbit to provide full global transmission.

Satellite Communication



Geosynchronous Orbit



PROGRESS PROJECT